

Measuring the Quantum Share of Mutation: A Heavy-Water Mutation-Accumulation Experiment

Harsh Goyal

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Abstract

UTOE predicts that a fraction f_q of spontaneous transition mutations arises from proton tunneling in DNA base pairs (Löwdin tautomers). Deuterium tunnels far less than hydrogen ($P_D \approx P_H^{\sqrt{2}}$), so growing cells in heavy water should selectively suppress that fraction. This document specifies a mutation-accumulation experiment in mismatch-repair-deficient *Escherichia coli* that measures f_q from the ratio of transitions to transversions across D₂O fractions and temperatures, and reports a full Python simulation of the experiment, including data generation, the pre-registered analysis and a power analysis. With 100 lines per condition (900 lines, 3,000 generations each), a 10% quantum share is detected with 85% power at a calibrated 5% false-positive rate. The simulation also shows a limitation the design must address: a classical isotope effect specific to transitions can mimic tunneling, and a temperature series alone cannot rule it out at this scale. We therefore pair the in vivo experiment with an in vitro polymerase kinetics test of the standard tunneling criteria.

Keywords: proton tunneling, heavy water, D₂O, mutation accumulation, kinetic isotope effect, quantum biology, experimental design, power analysis

1. Aim and hypothesis

Question. What fraction f_q of spontaneous transition mutations arises from proton tunneling between paired DNA bases?

Hypothesis (H1). $f_q > 0$: some transitions arise when a proton tunnels across a base-pair hydrogen bond, creating a rare tautomer that mispairs during replication [1, 2].

Null (H0). $f_q = 0$: transitions arise only from classical processes.

Core prediction. Replacing hydrogen with deuterium suppresses tunneling by a factor $\rho = P_D/P_H \approx P_H^{\sqrt{2}-1}$, which is close to zero for any realistic barrier. It does not suppress classical processes by anywhere near as much. If a fraction x of the relevant protons is deuterium, the quantum contribution scales as $(1 - x + x\rho)$, so **the share of transitions should drop by about $f_q \cdot x$** relative to a control class of mutations that tautomers do not produce.

2. Why the design works

Three facts make this experiment feasible.

1. **Heavy water deuterates exactly the right protons.** The protons in base-pair hydrogen bonds (the imino and amino N–H protons) exchange with the surrounding water within milliseconds to

seconds. Cells grown in a medium with D₂O fraction x therefore carry deuterium at about a fraction x of those positions, at the moment of replication.

2. **Tautomers cause transitions, not transversions.** A rare tautomer of G pairs with T, and a rare tautomer of A pairs with C. Both lead to **transitions** ($G \cdot C \leftrightarrow A \cdot T$). **Transversions** come from other chemistry (oxidative damage, other mispairs) and serve as an internal control.
3. **Ratios cancel the nuisances.** Heavy water slows growth and may change overall mutation rates through ordinary chemistry: polymerase speed, proofreading and nucleotide pools. These effects act on both mutation classes. They cancel in the ratio of transitions to transversions, as do any errors in counting generations. Within one condition (x, T) :

$$\text{odds}(x, T) = \frac{\text{transitions}}{\text{transversions}} = \theta_T \cdot h(x), \quad h(x) = 1 - f_T x (1 - \rho).$$

The baseline θ_T is free for each temperature, so the only quantity linking the conditions is f_T , the quantum share at temperature T .

3. Materials and design

3.1 Organism and strain

Escherichia coli K-12 MG1655 with a deletion of **mutL**, a mismatch-repair gene. In wild-type cells, mismatch repair removes most replication errors, including tautomer-induced mispairs, before they become mutations. Without it, the replication error spectrum is visible directly. MutL-deficient mutation-accumulation lines are dominated by transitions and have been characterized by whole-genome sequencing [3].

Expected rates in H₂O at 37 °C (used in the simulation): about 0.035 transitions and 0.002 transversions per genome per generation.

3.2 Conditions

Factor	Levels	Reason
D ₂ O fraction x	0, 0.5, 0.9	The ends carry most of the information; 0.5 tests the shape of the response
Temperature	25, 37, 42 °C	Tunneling is temperature-independent; classical chemistry is not
Lines per condition	100	From the power analysis below
Generations per line	3,000	Gives about 600 transversions per condition, the limiting count

That is $3 \times 3 = 9$ conditions and **900 lines in total**. Media and agar are prepared with the stated fraction of 99.9% D₂O. D₂O is the main consumable cost, so plate volume should be minimized (small plates or multi-well formats).

3.3 Procedure

1. **Founding.** Streak the ancestral mutL strain from a single colony. Freeze the ancestor, then found every line from one colony of it.
2. **Bottlenecking.** Every day, pick one random colony per line (the colony nearest a pre-marked point, to avoid choosing by size) and streak it onto a fresh plate of the same condition. Single-cell bottlenecks make natural selection negligible, so almost every mutation is kept regardless of its effect.
3. **Counting generations.** About every 10 transfers, suspend whole colonies from a sample of lines and count cells. Generations per transfer are $\log_2(\text{cells per colony})$, typically 25–28. Heavy water and low temperature slow growth, so they need more days but not more transfers per generation. Generation counts only affect the secondary temperature analysis, because they cancel in the primary ratio.
4. **Freezing.** Freeze every line every 20 transfers as a backup.
5. **Duration.** About 110–120 transfers reach 3,000 generations: roughly four months at 37 °C in H₂O, longer in D₂O and at 25 °C.
6. **Sequencing.** Extract genomic DNA from the final population of each line and the ancestor. Sequence with Illumina at 50× coverage or more, call variants against the ancestor (for example with *breseq*), and discard any variant present in more than one line (ancestral or contamination).
7. **Classification.** Classify each single-base substitution as a transition or a transversion. Record its sequence context for secondary analyses. Indels are excluded from the primary analysis.

3.4 Pre-registered analysis

- **Primary test:** one f_q shared across temperatures, with a free baseline θ_T at each temperature, fitted by maximum likelihood on the binomial split of mutations into transitions and transversions. H₀ ($f_q = 0$) is tested with a one-sided signed-root likelihood-ratio statistic, $z = \text{sign}(\hat{f})\sqrt{2\Delta\ell/\hat{\phi}}$, where $\hat{\phi}$ is the overdispersion estimated from line-level Pearson residuals. Reject H₀ at $z > 1.645$.
- **Secondary, temperature check:** fit f_T separately at each temperature. Because tunneling does not speed up with temperature while classical mutation does, a real quantum share must **fall** with temperature:

$$f_T = \frac{F}{F + (1 - F) e^{-\frac{E_a}{k_B} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}} \right)}},$$

where E_a is measured from the absolute transversion rates in H₂O. - **Reported in all cases:** $\hat{f}_q$ with a 95% confidence interval, so that a null result still sets an **upper bound** on the quantum share.

4. Simulation

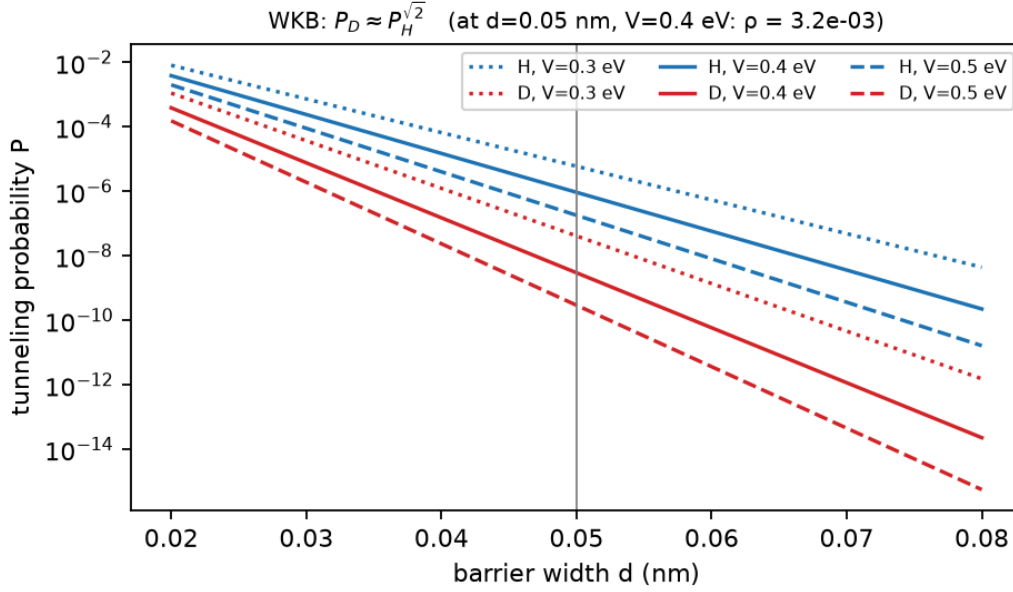
The whole experiment is simulated in `heavy_water/simulate.py`. The program:

1. computes the tunneling probabilities P_H and P_D from the WKB formula;
2. generates mutation counts for every line from a model with classical and quantum components, including Arrhenius temperature dependence for the classical part, a classical isotope effect shared

- by all mutations, 10% line-to-line rate variation and 3% error in generation counts;
- 3. runs the pre-registered analysis on the simulated data; and
- 4. repeats the whole experiment 300 times per design to estimate power and false-positive rate.

All biological parameters are illustrative, chosen from the published range for mutL *E. coli* [3]. The barrier height (0.40 eV) and width (0.05 nm) are illustrative too. The design is insensitive to them, because ρ is close to zero for any realistic barrier.

4.1 Tunneling physics

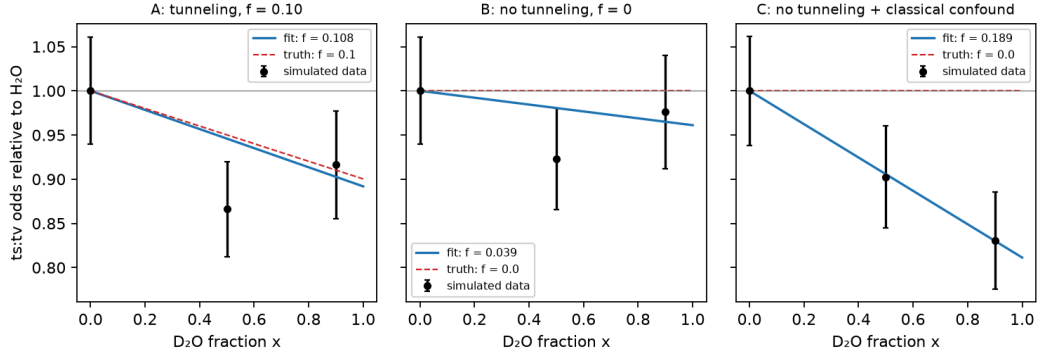


For the reference barrier, $P_H = 9.3 \times 10^{-7}$ and $P_D = 3.0 \times 10^{-9}$, so $\rho = 3.2 \times 10^{-3}$: deuterium tunnels about 300 times less often. At $x = 0.9$, the tunneling contribution therefore falls to about 10% of its H_2O value.

4.2 Three simulated worlds

Each scenario is one complete simulated run of the 900-line experiment.

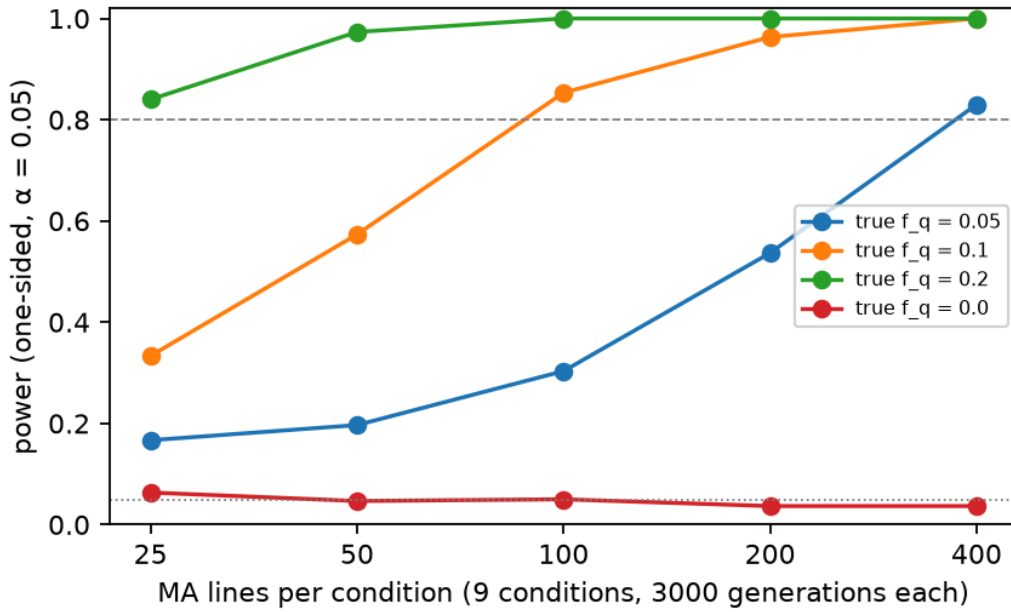
Scenario	Truth	Estimate $\hat{f}_q$	z	Verdict
A: tunneling	$f_q = 0.10$	0.114 ± 0.039	2.80	Detected, close to the truth
B: no tunneling	$f_q = 0$	-0.031 ± 0.044	-0.71	Correctly not detected
C: classical confound	$f_q = 0$, plus a transition-specific classical isotope effect	0.160 ± 0.039	3.76	False detection



4.3 Power

Lines per condition	25	50	100	200	400
$f_q = 0.20$	0.84	0.97	1.00	1.00	1.00
$f_q = 0.10$	0.33	0.57	0.85	0.96	1.00
$f_q = 0.05$	0.17	0.20	0.30	0.54	0.83
$f_q = 0$ (false positives)	0.06	0.05	0.05	0.04	0.04

300 simulated experiments per cell; one-sided test at the 5% level.



The test is **calibrated**: when there is no tunneling, it reports a detection about 5% of the time, as it should. With 100 lines per condition, a 10% quantum share is detected 85% of the time and a 20% share almost always. A 5% share needs about 400 lines per condition. The limiting factor throughout is the number of transversions, the control class, which are about 17 times rarer than transitions in mutL cells.

An earlier version of the design, with five D₂O levels, 40 lines and 1,000 generations, had an uncertainty of about ± 0.10 , as large as the effect itself. Concentrating lines at the ends of the D₂O range and running

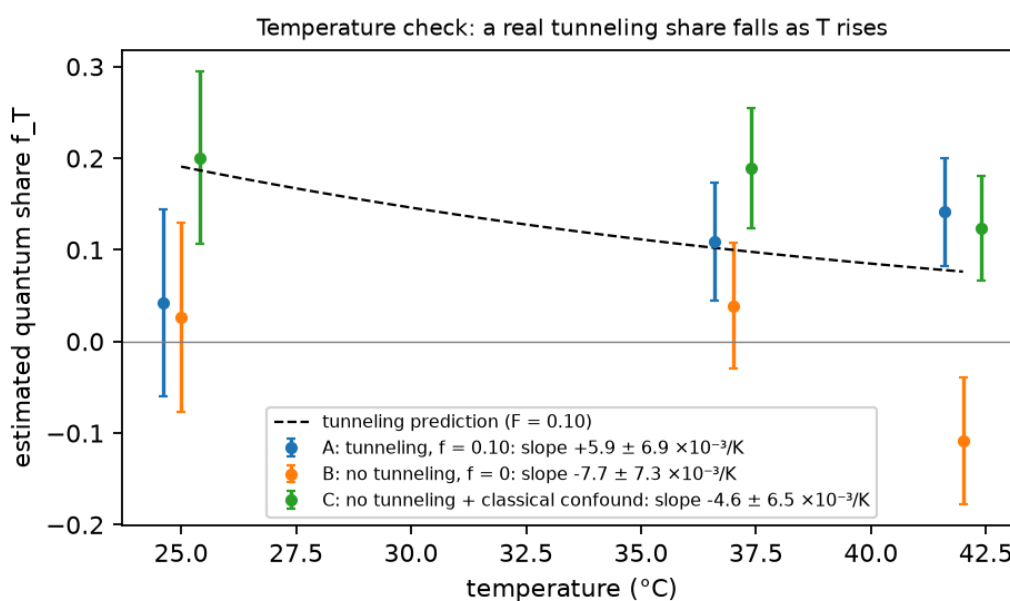
three times as many generations cut the uncertainty by more than half.

4.4 The confound, and why temperature alone cannot catch it

Scenario C is the main threat to the experiment. If heavy water slows the **classical** chemistry of transitions more than that of transversions (for example through a zero-point-energy effect on over-the-barrier tautomerization), the transition share drops even with no tunneling at all. The primary test cannot tell the two apart.

In principle, temperature can. A tunneling share falls as temperature rises; a classical one stays roughly flat. The simulation shows that the signal is too weak at this scale:

Scenario	Observed slope df_T/dT ($\times 10^{-3}/K$)	Tunneling would predict
A: tunneling	$+5.9 \pm 6.9$	-5.9
B: no tunneling	-7.7 ± 7.3	0
C: classical confound	-4.6 ± 6.5	-9.3



The uncertainty (about $\pm 7 \times 10^{-3}/K$) is as large as the effect, so the temperature arm cannot separate scenario A from scenario C. The 25–42 °C range is too narrow, because living *E. coli* cannot be grown much outside it.

4.5 Resolving the confound: an in vitro companion experiment

The established way to identify tunneling in chemistry and enzymology is the **temperature dependence of the kinetic isotope effect** over a wide range, which is easy in a test tube and impossible in living cells. The companion experiment:

1. Measure single-turnover misincorporation kinetics (G·T and A·C mispair formation) for purified *E. coli* DNA polymerase III, or a model polymerase, in H₂O and D₂O buffers from 5 to 50 °C.

2. Fit Arrhenius plots for both isotopes. Tunneling is indicated by the standard criteria: a large isotope effect (above about 7 at 25 °C), an isotope-dependent activation energy difference larger than the zero-point energy limit, and a ratio of Arrhenius prefactors A_H/A_D outside the semiclassical range of about 0.7–1.2 [4].
3. Run the same assay on transversion-forming mispairs as a control.

5. Interpreting the outcomes

In vivo result	In vitro tunneling criteria	Conclusion
No transition-specific drop ($z < 1.645$)	Any	f_q is below the measured 95% upper bound; tunneling is not a major mutation source
Significant drop	Met	Evidence that a fraction $\hat{f}_q$ of mutation is quantum in origin
Significant drop	Not met	A classical, transition-specific isotope effect; no evidence for tunneling

Every outcome is informative. Even a null result is new: the first direct experimental upper bound on how much of spontaneous mutation comes from proton tunneling.

6. Limitations

- **Illustrative parameters.** The rates, activation energy and isotope-effect energies in the simulation are plausible but not measured for these exact conditions. A small pilot (for example 20 lines per condition for 500 generations) should be used to re-estimate them and re-run the power analysis before the full experiment.
- **Proton inventory.** The model assumes the tunneling contribution falls linearly with x . If tautomers form by concerted transfer of two protons, the dependence may be closer to $(1 - x)^2$. The $x = 0.5$ level allows this to be checked, and the analysis can be refitted with either form.
- **Heavy water as a stressor.** High D₂O fractions slow growth and may trigger stress responses that change mutation rates. The transversion control absorbs effects shared by both classes, but stress-induced mutagenesis with a transition bias would act as another confound. It is addressed by the same in vitro companion experiment.
- **Scale and cost.** 900 lines for four months or more, and the cost of D₂O media, make this a substantial project. The power analysis shows the scale is needed; a smaller experiment could only detect a quantum share of 20% or more.

7. Reproducing the simulation

```
python heavy_water/simulate.py # full run, a few minutes
python heavy_water/simulate.py --quick # fewer power-analysis repeats
```

The program writes the four figures and `results.json` (all numbers above) to the `heavy_water/` folder. It uses NumPy, SciPy and Matplotlib.

References

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